D222 with LoRMA NER tuning

CS781 Phase 2 legal named entity recognition | source-audited report

Task and evidence

Predict exact original-character entity spans in 12 legal categories. The source archive holds the actual code, configuration, run record, selected weights, and prediction ZIP. This report was produced after training from those records; it is not an executed historical notebook.

Architecture

The same D222 decoder is adapted by rank-8 multiplicative low-rank matrices, training a small fraction of its weights. Character BiGRU, boundary logits, and CRF form the NER head.

Path	Legal text -> Own byte BPE -> 24-layer D222 GQA -> Rank-8 LoRMA+
Output	Character BiGRU -> CRF + boundaries -> Span export

Training and validation

Method: LoRMA+, rank 8. This export's archived held-out strict macro-F1 is 0.427679 on the 12-document local validation split. Its selected checkpoint and prediction ZIP hashes are in evidence/run_summary.json. The official test set has 35 document IDs; no hidden Codabench score is established by this archive.

Implementation map

Key original files: legal_d/model.py, legal_d/ner.py, legal_d/ner_train.py. The companion notebook explains the tensor route and reads archived evidence without presenting invented execution outputs.

Reproducibility boundary

The package includes exact source snapshots and small run records, plus the archived prediction ZIP. Large model weights, tokenizers, and course data stay in the separate complete archive. Original machine paths in config files need remapping for a new environment. The consensus rule was selected on validation; 100M/200M D222 exports were completed after the original deadline where marked.

Interpretation

Model variants change either pretrained backbone, parameter-update method, NER head, or pretrained checkpoint. A score difference does not isolate one factor unless all other choices match. The local held-out set is small, so its ranking should not be described as a hidden-test result.

Recorded model and adaptation

The source and run records together identify the following settings. Counts include the Phase 2 NER head. The method changes which backbone parameters are updated; it does not change the original Phase 1 tokenizer during this run.

Field	Archived value
Base family	D222
Adaptation	LoRMA+, rank 8
Total parameters	225852926
Trainable parameters	3773054
Backbone LR	0.0002
Head LR	0.0005
Pretrained SHA-256	see run.json…
Selected NER SHA-256	86996d831f79031b862e…

Training signal and exact-span evaluation

For the common NER runner, the character tagger minimizes weighted cross-entropy over 25 BIO categories; A0 also matches its verified B1 teacher with temperature-2 KL. D222 adds CRF negative log-likelihood and a weighted boundary objective. HeaderCRF combines cross-entropy, CRF, and boundary losses. The validation scorer counts a prediction as correct only when the start, end, and label all match the gold span; macro-F1 averages the 12 per-label F1 values equally.

Observed optimization trace

These five entries are sampled from the archived training log; loss is a per-batch training value, not held-out macro-F1. Its decline alone cannot establish improved entity extraction.

Optimizer step	Epoch	Logged batch loss
1	1	3.1815
330	2	0.2894
670	5	0.3117
1000	6	0.3109
1330	8	0.2464

Label-level held-out scores

The following F1 values come from the archived selected validation prediction. Low-frequency labels are sensitive to a few documents, so read them beside their support and error counts in evidence/selection.json.

Entity label	F1	TP	FP	FN
APP	0.098	10	86	99
RESP	0.208	22	89	79
A.COUNSEL	0.420	21	34	24

Entity label	F1	TP	FP	FN
R.COUNSEL	0.333	11	24	20
JUDGE	0.597	23	19	12
WIT	0.000	0	4	21
AUTH	0.306	83	255	122
COURT	0.743	156	54	54
STAT	0.704	238	94	106
PREC	0.246	64	195	198
DATE	0.941	319	27	13
CASENO	0.537	80	80	58

Verification and reproducibility

The prediction ZIP has a single root prediction.json, passed CRC, and contains exactly 35 official test IDs. Its SHA-256 was compared with the completed run record. Each copied code file has a SHA-256 listed in evidence/SOURCE_HASHES.json. Exact re-execution requires the original selected weights, tokenizer and course data preserved in the complete archive; this compact package does not embed them or assert that historical notebook output cells existed.